

Cartesian product of two sets.

Given sets A, B , the cartesian product of A and B is denoted by $A \times B$

$$A \times B = \{ \langle a, b \rangle : a \in A \text{ and } b \in B \}$$

contains all ordered pairs where first element comes from A and second element comes from B .

$$A = \{1, 2, 3\} \quad B = \{\Delta, \star\}$$

$$A \times B = \{(1, \Delta), (1, \star), (2, \Delta), (2, \star), (3, \Delta), (3, \star)\}$$

$$B \times A \neq A \times B, \text{ unless } A = B$$

Theorem and proofs

Theorem - A true mathematical statement.

Example: If x, y is rational, then xy is rational.

- A proof is a convincing logical argument that the statement is true.

- Given a universal statement, we can disprove it using counter example.

Ex: All swans are white

Cannot disprove
by a counter example.

there exist at least one green swan

All unicorns are Red

Ex: If $x \in \mathbb{N} \wedge (x \bmod 2 = 0)$, then
 x is a power of 2.

disprove by counter example

Let $x = 6$

$x \in \mathbb{N} \wedge (x \bmod 2 = 0)$ but

6 is not a power of 2

clearly, the claim does not hold.

Implications and If and only if
($\Rightarrow$) ($\Leftrightarrow$)

$$p \Rightarrow q$$

If moon is made out of
cheese then $2+2=7$

$$p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$$

Proof methods:

1. Direct proofs
2. proof by contradiction
3. proof by induction
4. proof by construction
($\neq$ by example)

Direct proof

Given a graph $G = (V, E)$, let $\deg(v)$ be the # of edges incident to v .

$\sum_{v \in V} \deg(v)$ is even

Proof: when counting degrees, each edge $\{u, v\}$ is counted twice and once for $\deg(v)$

$$\therefore \sum_{v \in V} \deg(v) = 2 \cdot |E| \quad \text{which is even.}$$

Example: Given Sets A and B

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

$\underbrace{\hspace{1.5cm}}_X \qquad \underbrace{\hspace{1.5cm}}_Y$

$$X = Y$$

$$\frac{X \subseteq Y \wedge Y \subseteq X}{\text{①} \qquad \text{②}}$$

①: WTS $\overline{A \cup B} \subseteq \overline{A} \cap \overline{B}$

If $x \in \overline{A \cup B}$, then $x \in \overline{A} \cap \overline{B}$

Let $x \in \overline{A \cup B}$

$$x \notin A \cup B$$

$$x \notin A \wedge x \notin B$$

$$x \in \overline{A} \wedge x \in \overline{B}$$

$$x \in \overline{A} \cap \overline{B}$$

②: WTS $\overline{A} \cap \overline{B} \subseteq \overline{A \cup B}$

Relations and functions

A function is a mapping between two sets, with set of rules.

formally function f from set A to B is written as

$f: A \longrightarrow B$ with following conditions.

1. $\forall a \in A : f(a)$ must be defined
2. $\forall a \in A : f(a)$ should produce unique element.
3. $\forall a \in A : f(a) \in B$

A - Domain

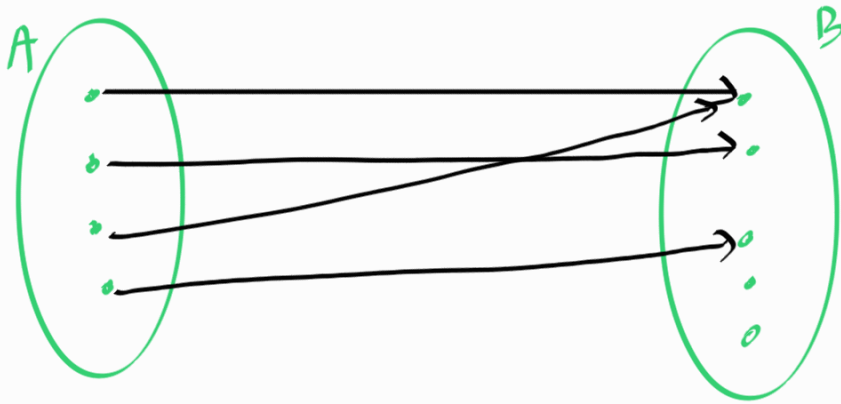
B - Co Domain

R - Range : range is the set of all outputs produced by the function

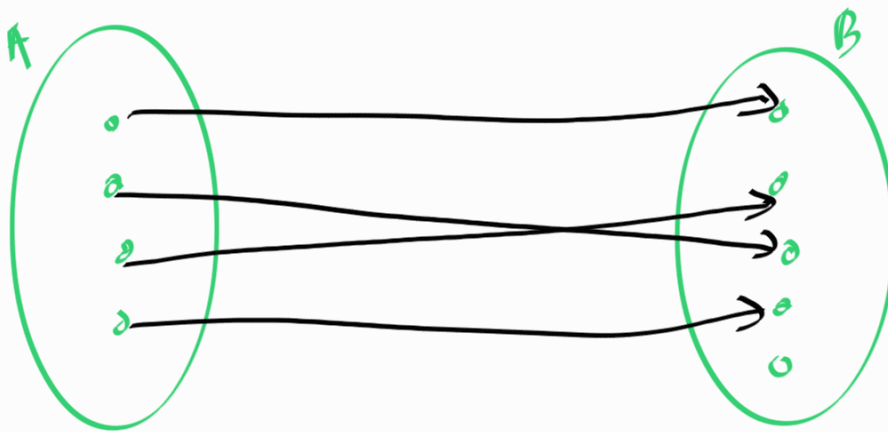
$$R \subseteq B$$

Injective functions (one-to-one)

$$1. \forall a_1, a_2 \in A : a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2)$$



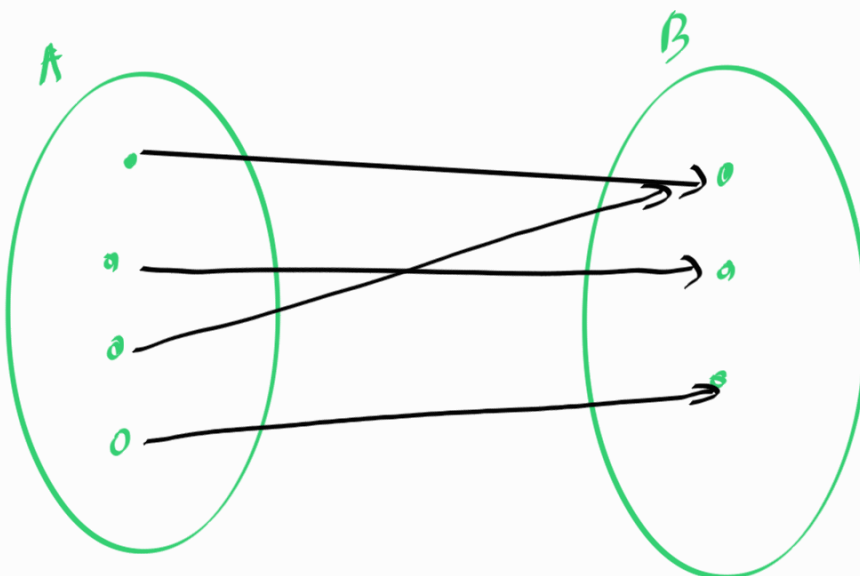
Not a
one-to-one
function



one-to-one

Surjective function (onto)

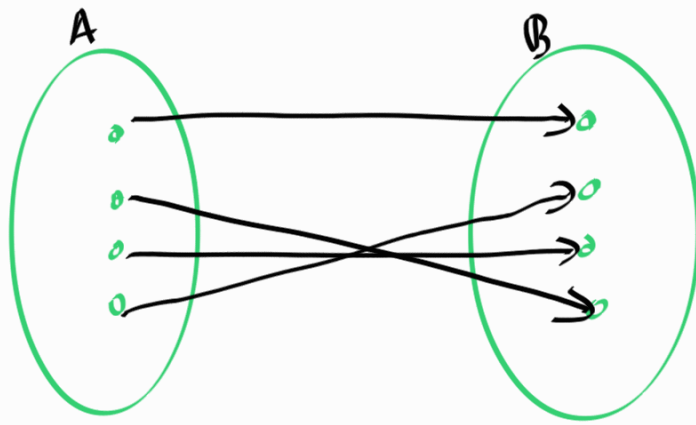
$$[\forall b \in B : [\exists a \in A : f(a) = b]]$$



onto
function

Bijection

Any function that is injective and surjective is called a bijection



Bijection

Predicates

Predicates / property is a function whose range is $\{\text{True}, \text{False}\}$

Ex: $\text{isEven} : \mathbb{Z} \longrightarrow \{\text{T}, \text{F}\}$

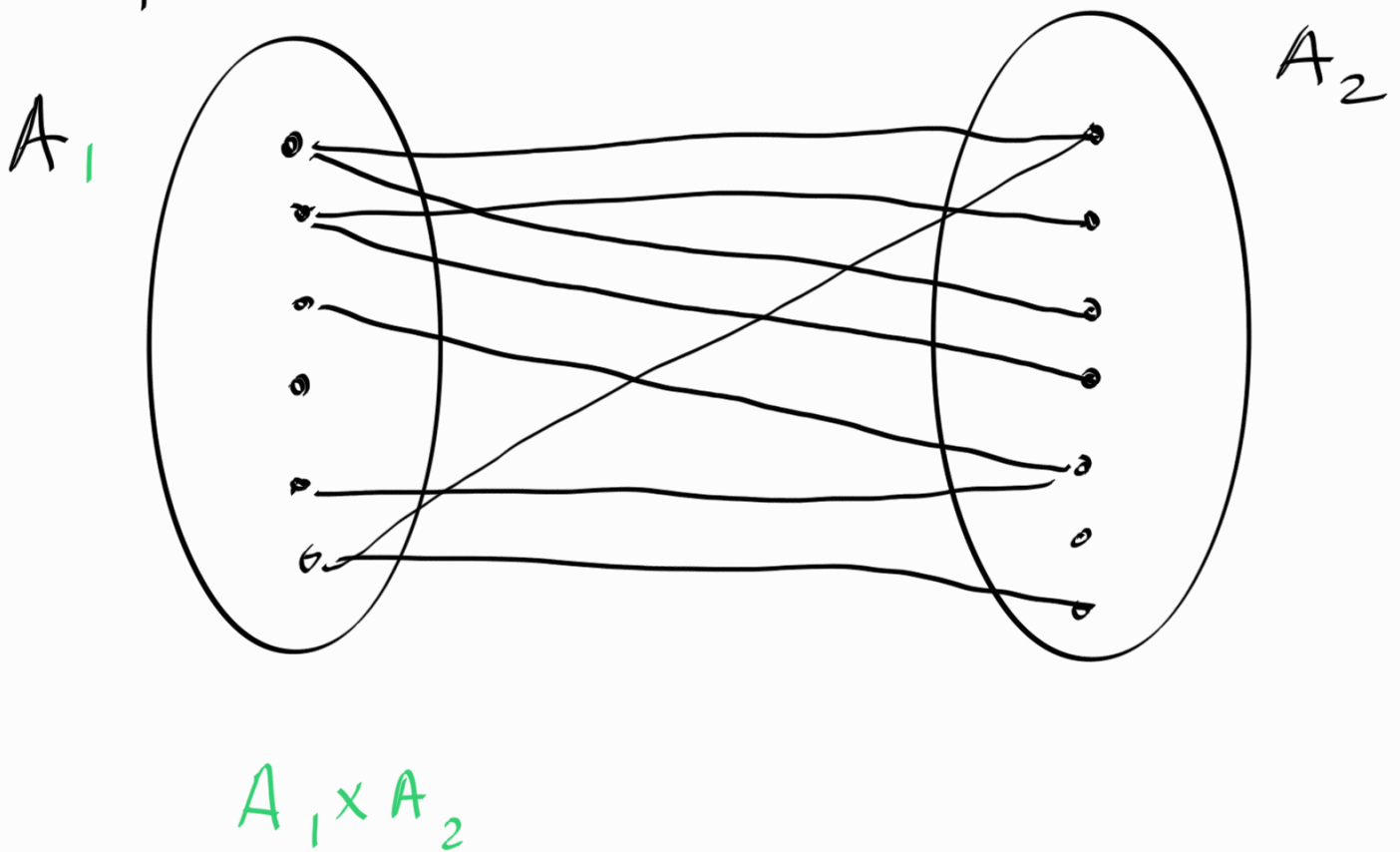
$$\text{isEven}(x) = \begin{cases} \text{True} & , \text{ if } x \text{ is even} \\ \text{False} & , \text{ if } x \text{ is odd} \end{cases}$$

Relations

k -ary relation R on
 k sets $A_1, A_2, A_3, \dots, A_k$ is
defined as follows

$$R \subseteq A_1 \times A_2 \times A_3 \times \dots \times A_k$$

Example $k=2$



In practice, we usually
look at $K=2$

Binary Relations

Special Binary Relations defined
on a single set:

Let A be a set

Let R be a subset of $A \times A$

$$R \subseteq A \times A$$

Equivalence Relation

Given a set A and a Binary

Relation R , $R \subseteq A \times A$

We say R is an equivalence relation

1. Reflexive

$$\forall x \in A: x R x$$

2. Symmetric

$$\forall x, y \in A: x R y \Rightarrow y R x$$

3. Transitive

$$\forall x, y, z \in A: x R y \wedge y R z \Rightarrow x R z$$

